FLicc: multi-gear length-based assessment in FLR

Concept, motivation, indicators, and management use

Henning Winker (FAO-NFIFM)

FLicc

Motivation

- `FLicc` is an **FLR-friendly TMB implementation and extension** of the multi-gear length-interval catch-curve framework developed in `fishblicc`.
- It retains the biological core of the original equilibrium model, but adds:
  - **fast penalized-likelihood estimation in TMB**
  - **multi-year fitting**
  - **FLR integration**
  - **equilibrium and indicator tools**
- The original motivation is simple: many real fisheries are **multi-gear**, but many length-based methods assume a **single gear with logistic selectivity**.

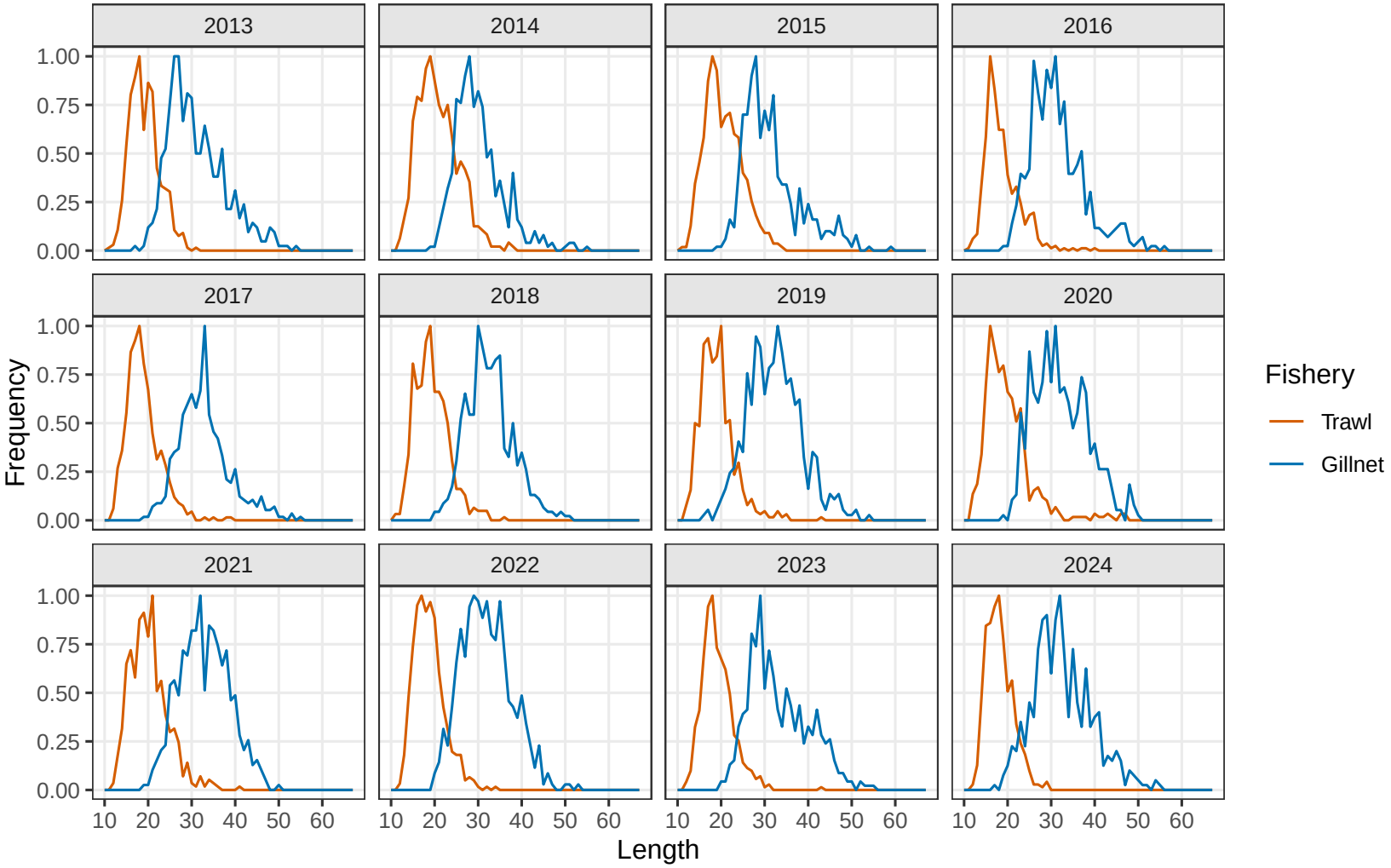
Why multi-gear matters

Long-lived species and mixed fleets

- In long-lived species such as **alfonsino**, mixed fleets often exploit **different parts of the size distribution**.
- This creates a core identifiability problem:
  - the declining right tail of the catch-at-length distribution may reflect
    1. real mortality, and/or
    2. the descending limb of gear selectivity.
- Joint fitting across gears helps separate:
  - **fishing mortality**, and
  - **selectivity shape**.

Example context: Alfonsino

Multi-gear length frequencies



- Alfonsino provides a useful example of a **slow-growing, long-lived, vulnerable species**.

Relationship to LBSPR

A reproducible and extensible bridge

- `FLicc` is closely related to **LBSPR**.
- Under the special case of:
  - a **single fleet**
  - **logistic selectivity**
  - a **multinomial likelihood**
- `FLicc` reproduces **LBSPR-type inference**, while extending the framework to more realistic fleet structures.

$$\frac{Z}{K} = \frac{M + F}{K}$$

- This length-structured per-recruit logic underpins both frameworks.

Link to fishblicc

From Stan to FLR/TMB

- `fishblicc` introduced a **Bayesian multi-gear catch-curve framework** for length data.
- `FLicc` keeps the core equilibrium interpretation, but adds:
  - **speed**
  - **joint multi-year fitting**
  - **FLR compatibility**
  - **equilibrium reference curves**
  - **indicator workflows**
- GitHub: **github.com/Henning-Winker/FLicc**

Core model idea

Catch-at-length under growth and mortality

$$c_i = \frac{F_i}{Z_i}(S_i - S_{i+1})$$

where:

- S_i = survival to the lower bound of length interval i
- F_i = fishing mortality at length
- $Z_i = M_i + F_i$ = total mortality at length

For gear j, catch per recruit is:

$$C_j = \sum_{i=1}^{N} \frac{F_{ij}}{Z_i}(S_i - S_{i+1})$$

Multi-gear decomposition

Shared population, gear-specific selectivity

$$Z_i = M_i + \sum_j F_j \, sel_{ij}$$

- All gears observe the **same underlying population**.
- Each gear samples that population through a **different selectivity function**.
- The joint likelihood constrains mortality and selectivity simultaneously.

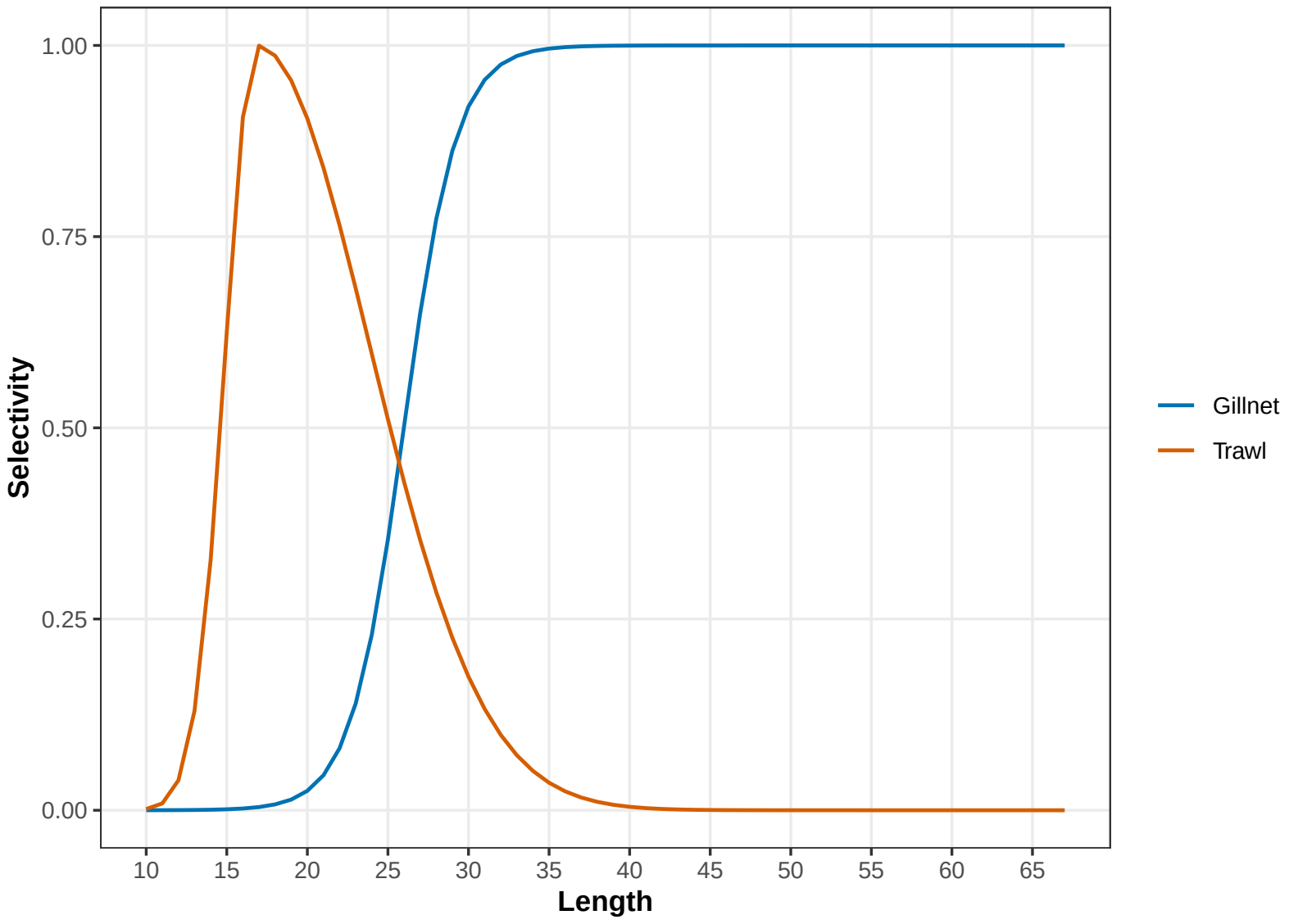
Selectivity options

Why flexibility matters

`FLicc` currently supports:

- **logistic**
- **normal**
- **double-sided normal**

This is especially important in fisheries where some gears are **dome-shaped**.



- A multi-gear framework is particularly useful when at least one gear still samples the larger fish.

Population and observation models

Population model + observation model separation

Default configuration:

```
settings = list(pop_model = "gtg", obs_model = "mn")
```

- `pop_model = "gtg"`: growth-type-group approximation
- `obs_model = "mn"`: multinomial observation model

This default is typically **much faster** than the original fishblicc-style gamma integration workflow while retaining the key length-based structure.

Population model options

GTG versus gamma

- **GTG (default)**
  - efficient recursion over growth-type groups
  - recommended for most applications
- **gamma**
  - closer to the original fishblicc integration approach
  - slower, but useful for replication / comparison

$$N_{i+1} = N_i \exp(-Z_i \Delta t_i)$$

- The model is still rooted in **equilibrium length structure under growth and mortality**.

Observation model options

Flexible composition likelihoods

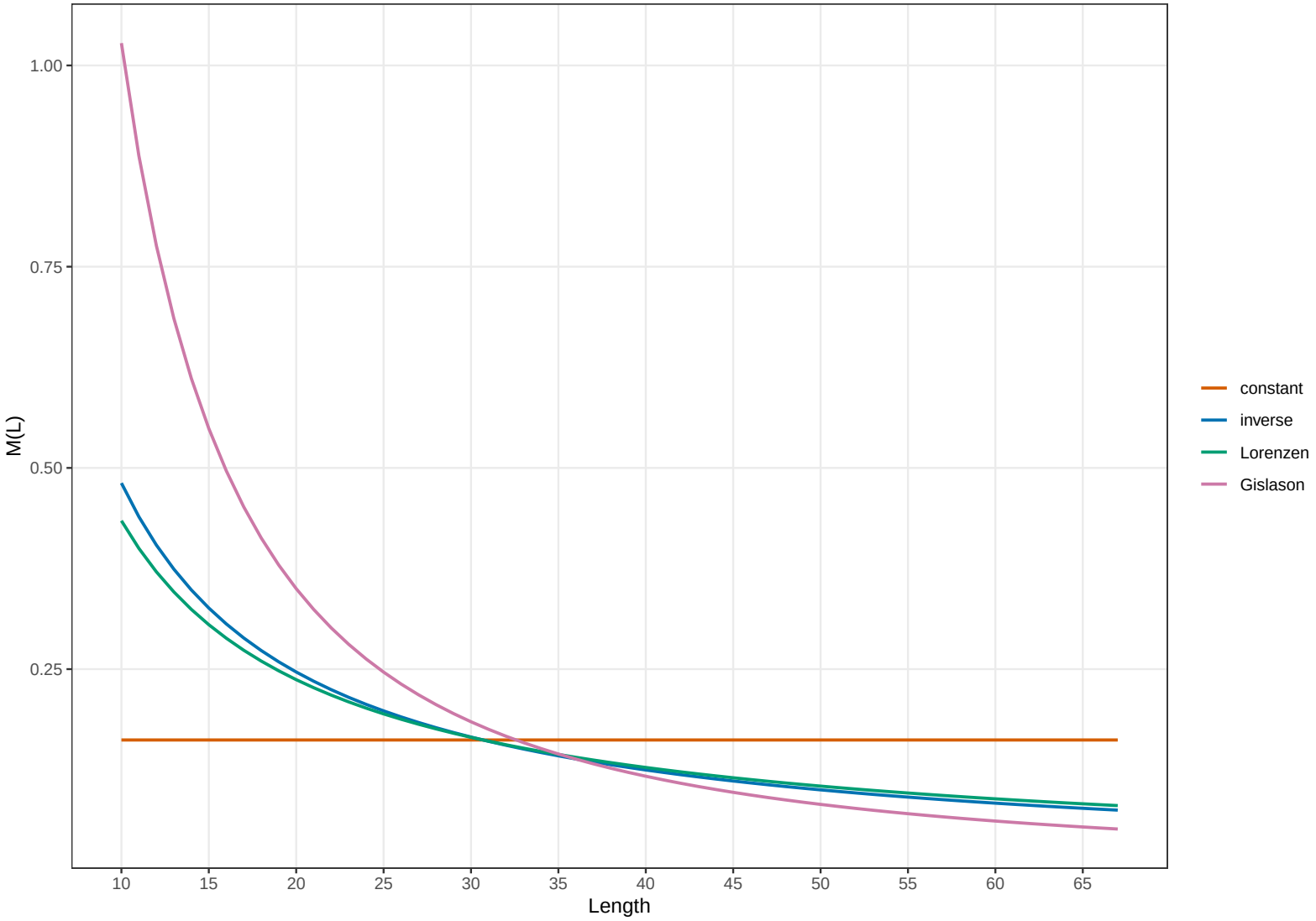
- **Multinomial**

$$\mathbf{y} \sim \text{Multinomial}(n, \boldsymbol{\pi})$$

- **Dirichlet-multinomial** for overdispersion
- **Negative binomial** for count-based flexibility
- In practice:
  - `mn` is fast and stable
  - `dm` is attractive when there is composition overdispersion
  - `nb` can be useful, but may be less stable

Life-history input

Build the FLStockLen object

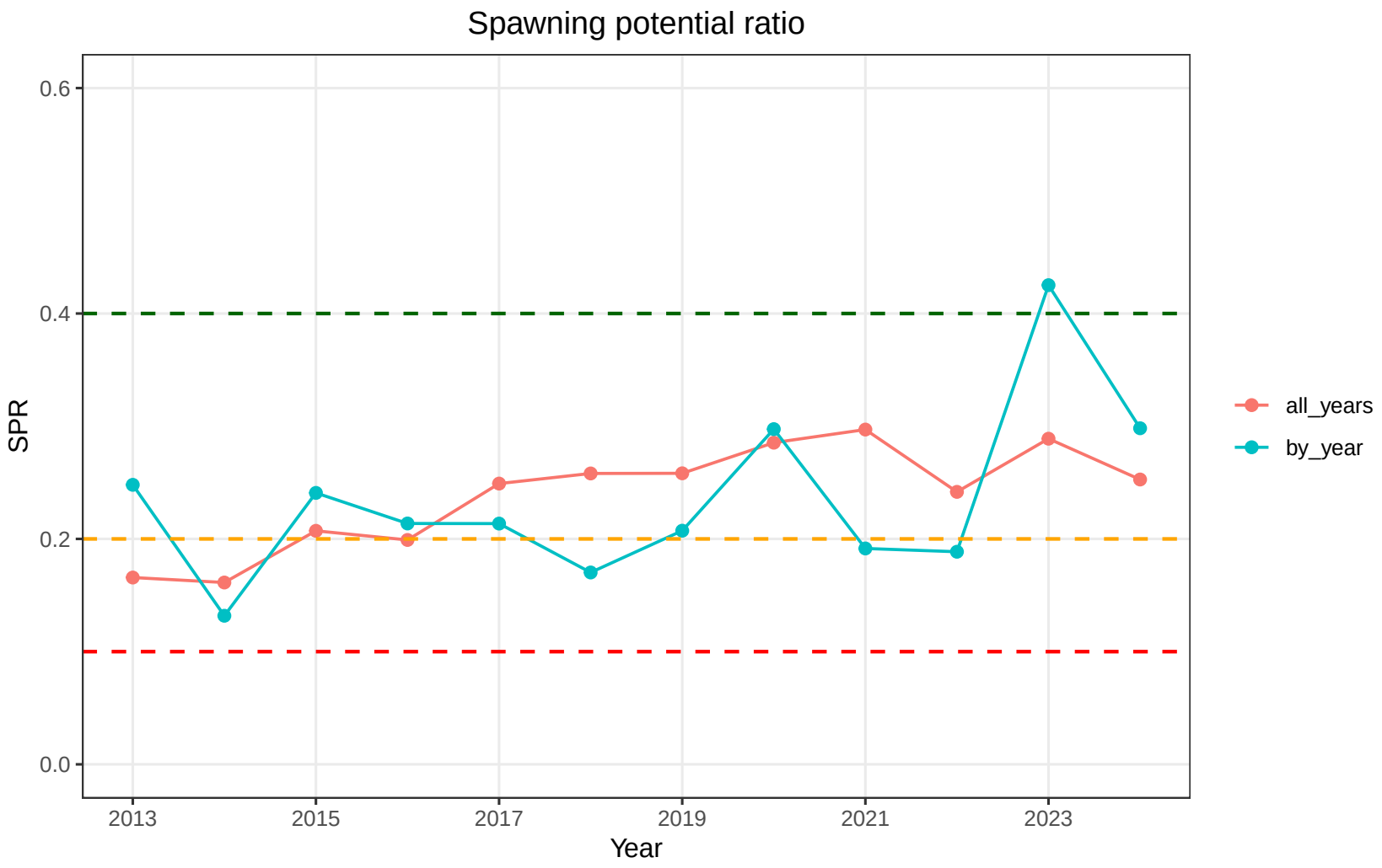


- `FLicc` uses an `FLStockLen`-style structure to hold:

Fitting across years

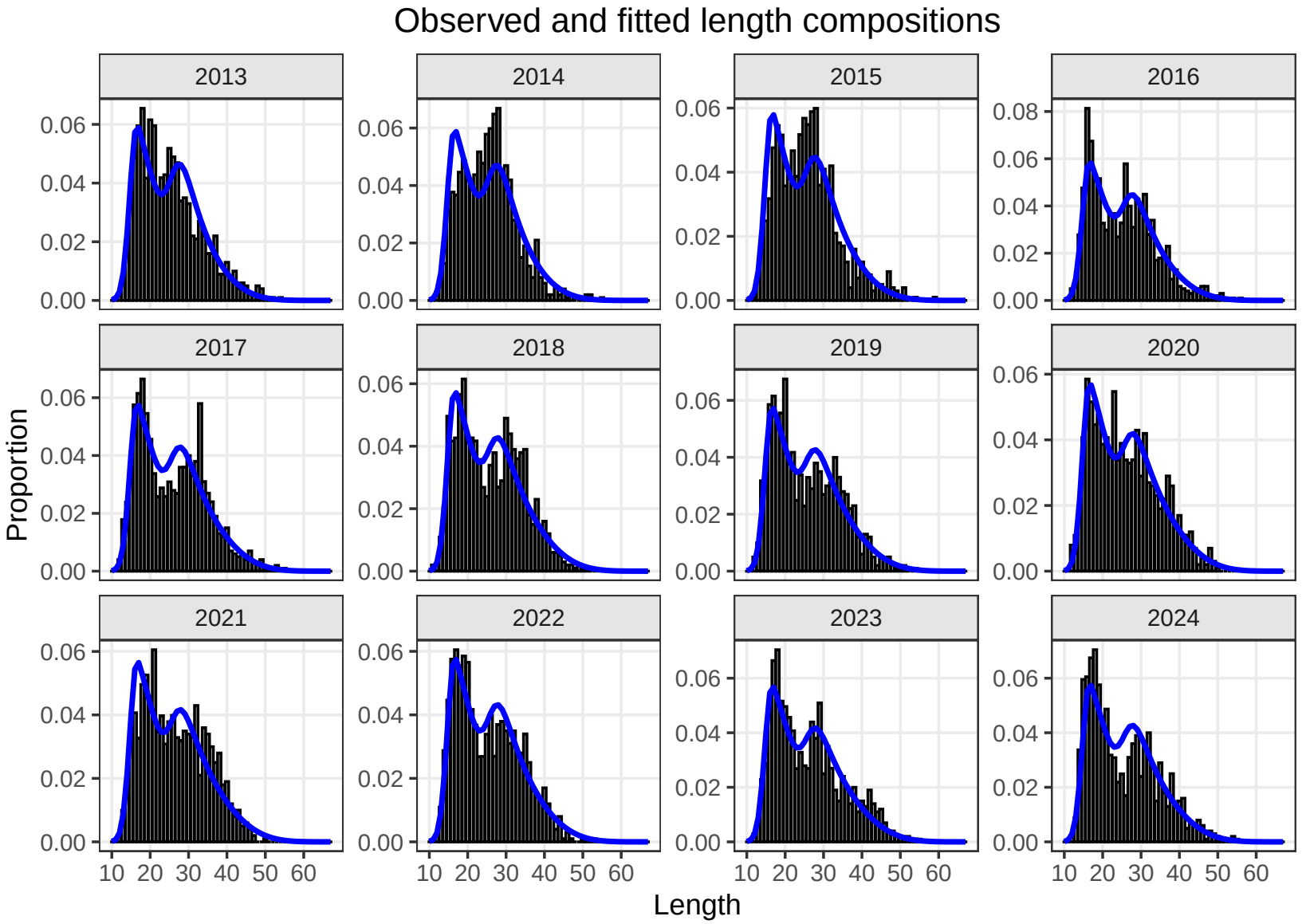
Borrowing strength across annual data

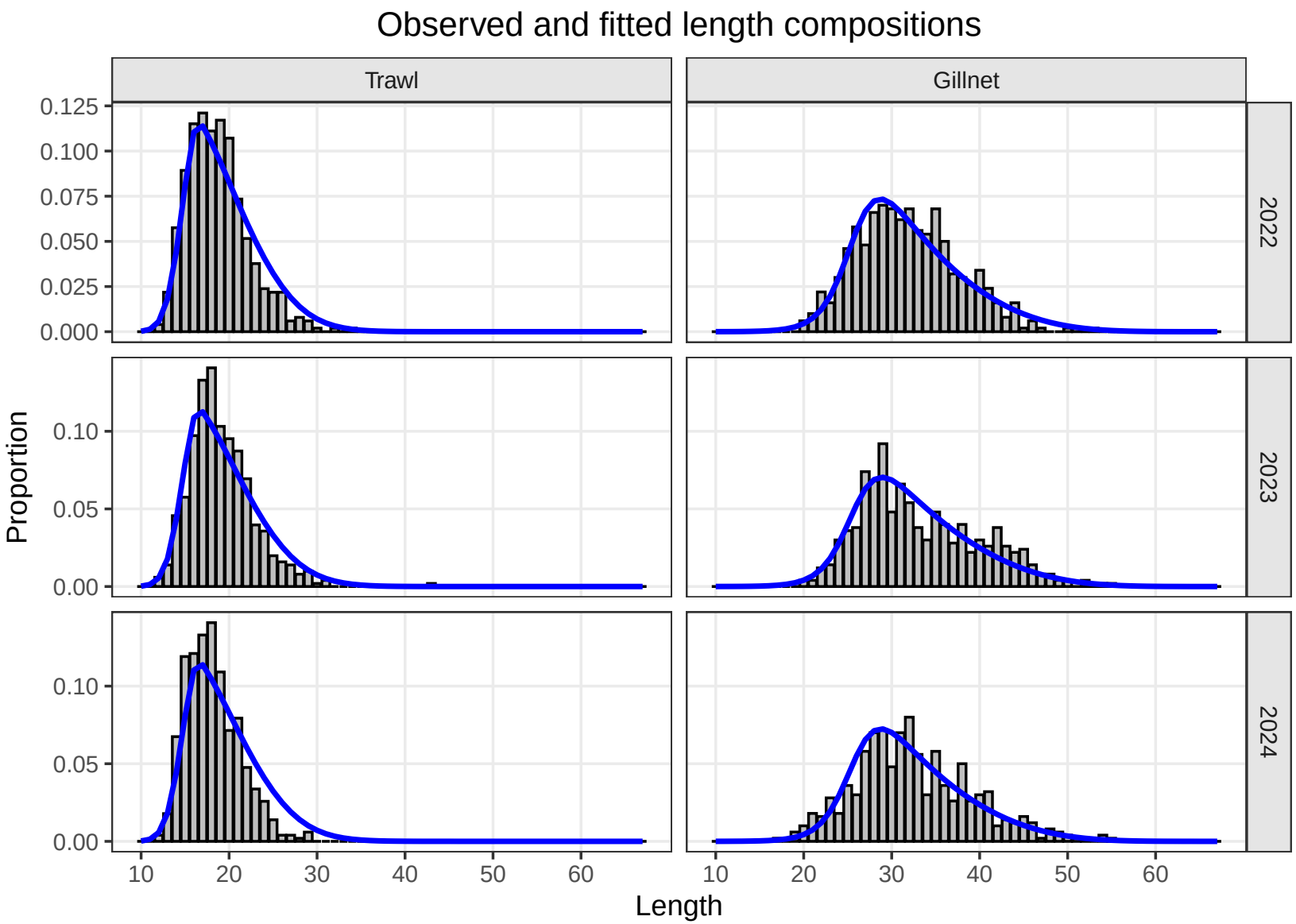
- `FLicc` can fit:
  - **all years jointly**, or
  - **each year separately**
- Multi-year fitting improves identifiability of:
  - life-history and selectivity parameters
  - year-specific fishing mortality


Spawning potential ratio
SPR
0.6
0.4
0.2
0.0
2013
2015
2017
2019
2021
2023
Year
all_years
by_year

Observed versus fitted length compositions

Model diagnostics

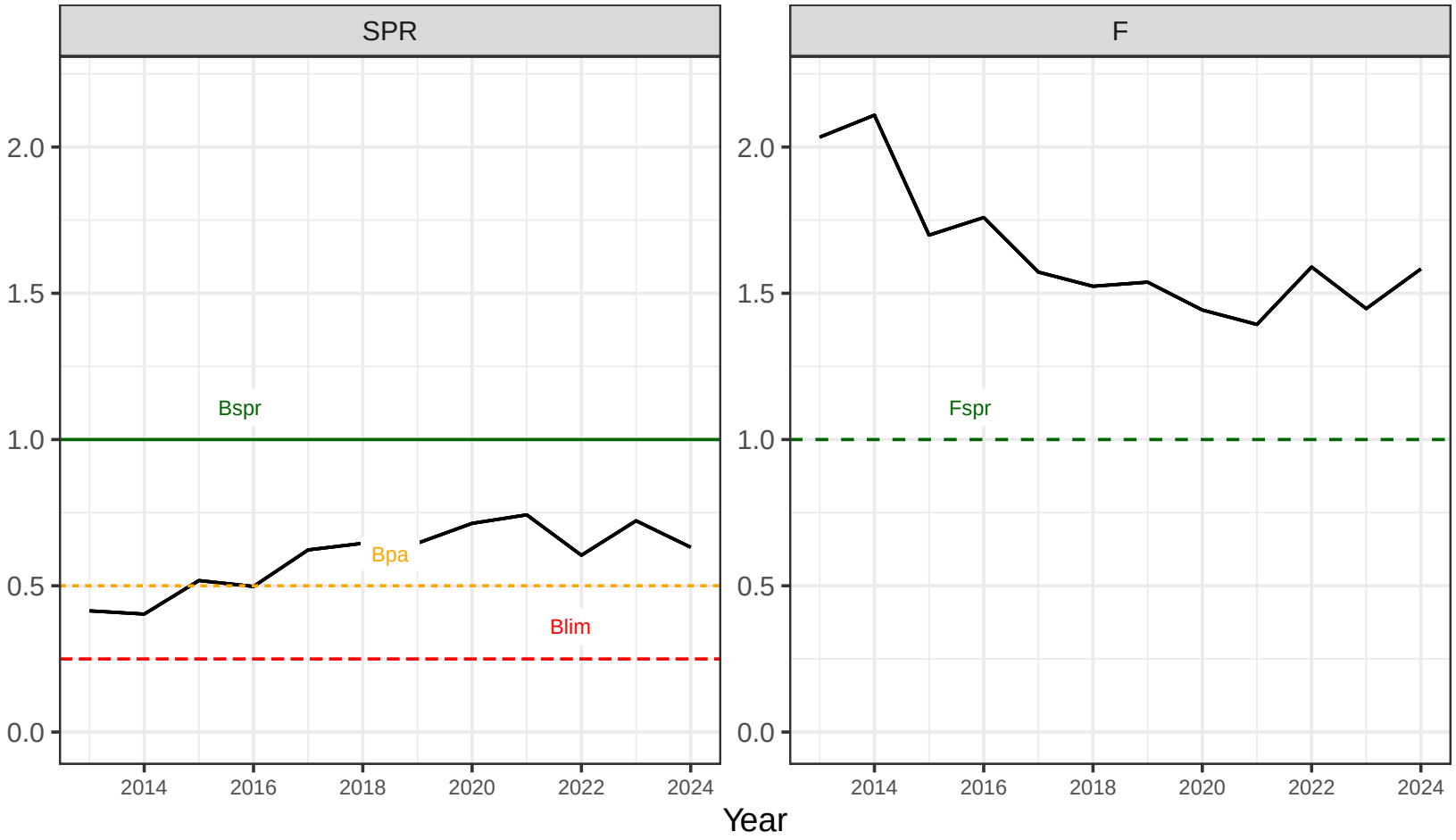




- Key diagnostic questions:
  - Are the dominant modes reproduced?
  - Is the upper tail represented?
  - Does each gear's fitted shape look plausible?

Status and reference points

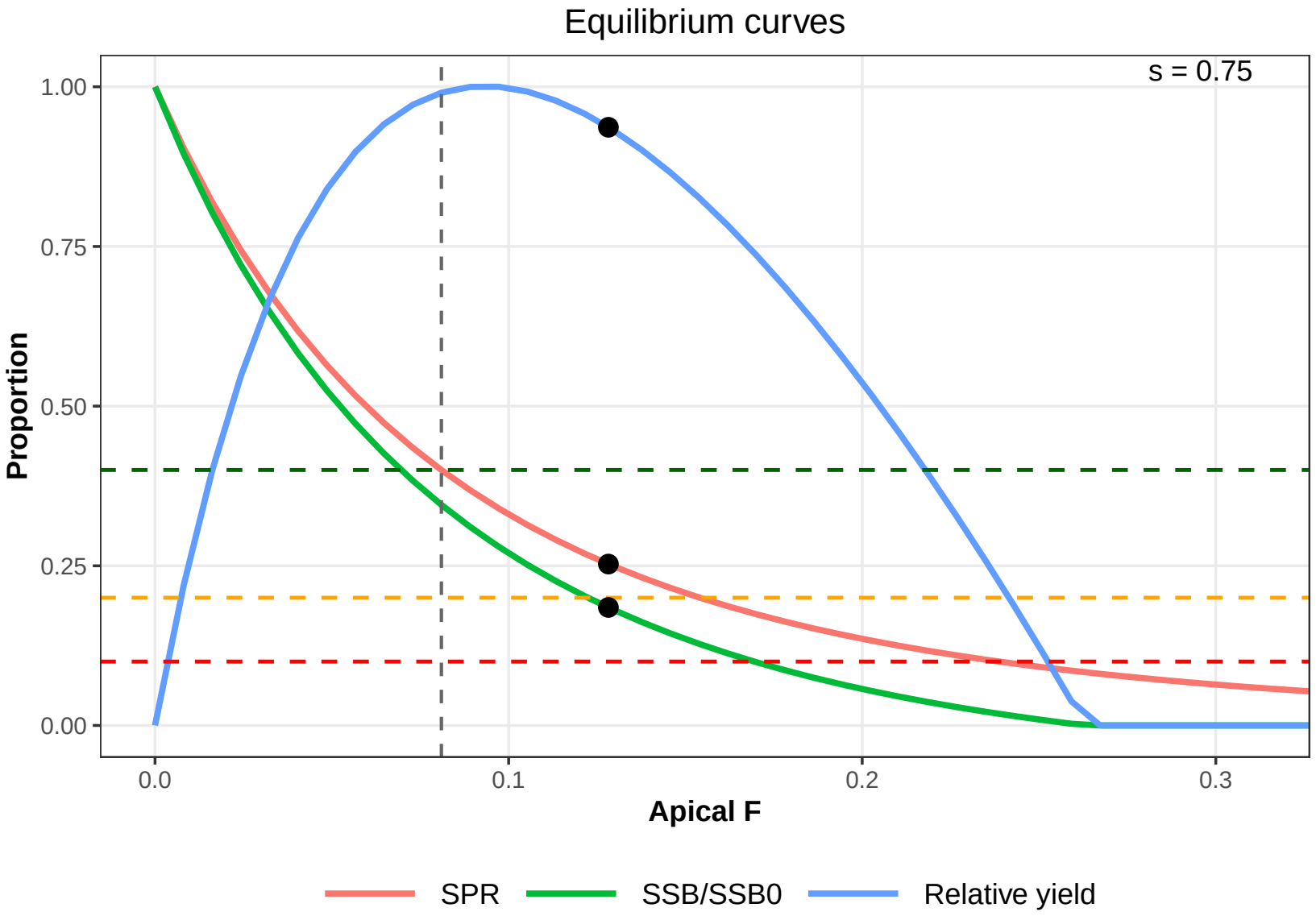
SPR as a core output



- `FLicc` provides annual estimates of **SPR**.
- These can be linked directly to **SPR40**, **SPR35**, or other target proxies.

Equilibrium curves

Per-recruit reference relationships



- `eqctl1er()` provides equilibrium relationships among:

What is new relative to fishblicc?

Practical extensions in FLicc

- **Multi-year estimation framework**
- **Random-walk structure in annual fishing mortality**
- **Alternative observation models**
- **FLR object outputs**
- **Indicator development**, including `LBIspr`
- **Fast TMB implementation**

These features make `FLicc` attractive both for **method development** and for **applied workflows**.

LBIspr motivation

Why a structure-based indicator?

- CPUE can be extremely valuable, but in many fisheries it may be:
  - unavailable
  - noisy
  - inconsistent through time
  - difficult to standardize across fleets
- Length data are often more widely available.

`LBIspr` was developed to provide a **structure-based index** that can:

- substitute for a relative abundance proxy when CPUE is weak or absent
- corroborate CPUE trends where CPUE exists

LBIspr concept

Observed structure versus expected target structure

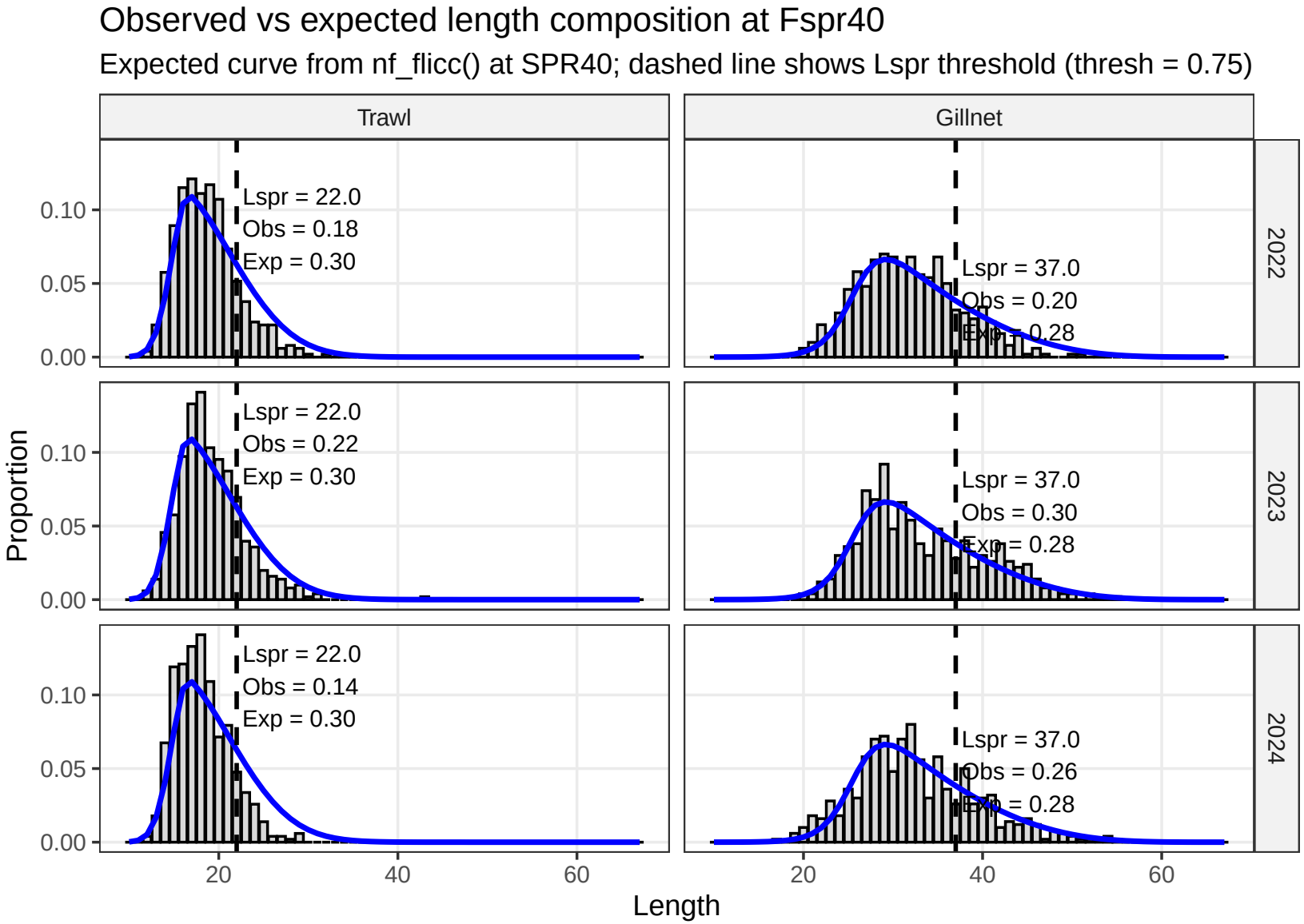
$$LBI_{SPRx,t} = \frac{P_t(L \geq L_{SPRx})}{P_{SPRx}}$$

where:

- L_{SPRx} is a large-fish threshold linked to a target SPR level
- $P_t(L \geq L_{SPRx})$ is the observed proportion above that threshold
- P_{SPRx} is the expected equilibrium proportion above that threshold under the target SPR

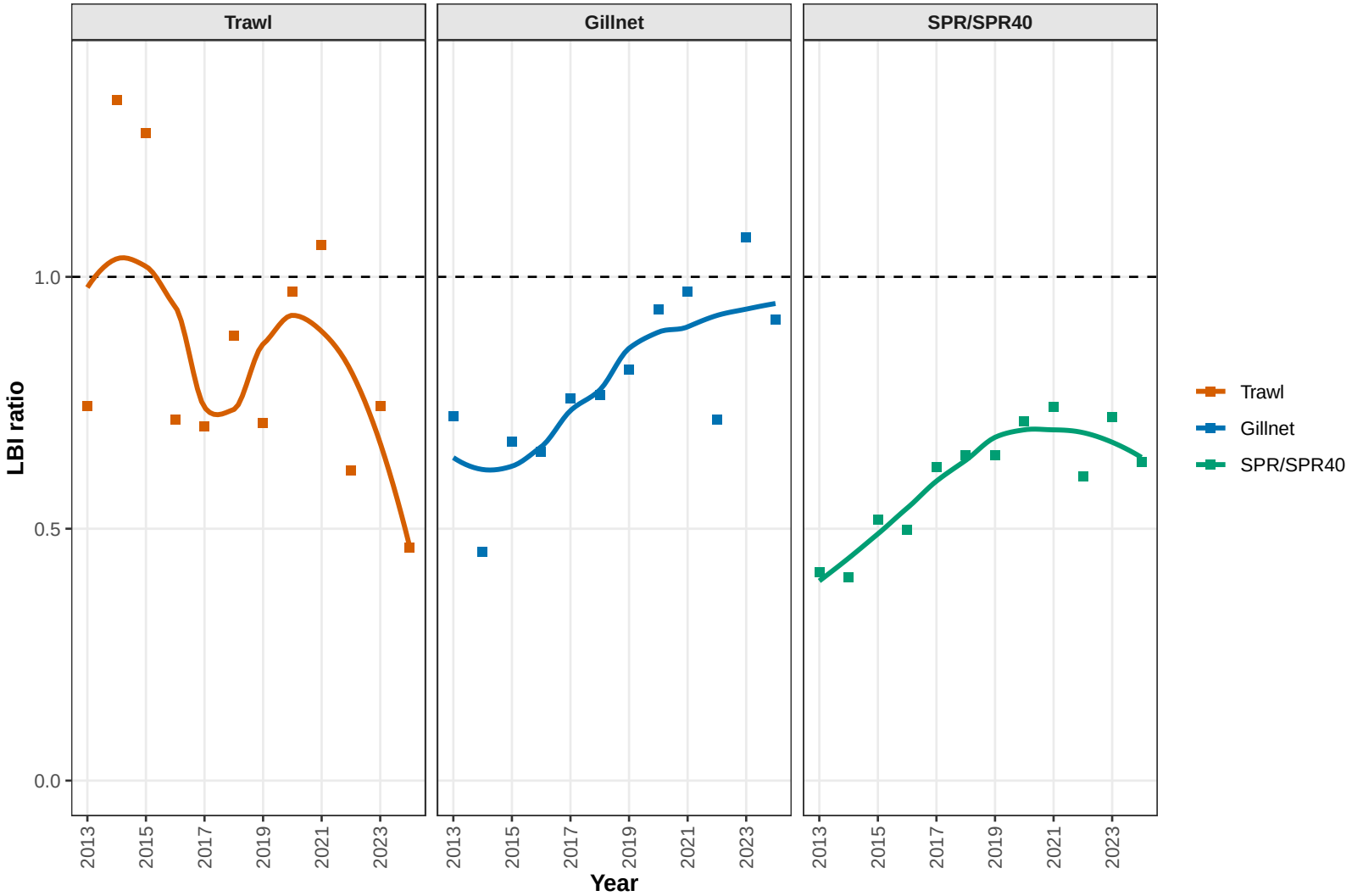
Interpretation:

- $< \mathbf{1}$: fewer large fish than expected under the target structure
- $\sim \mathbf{1}$: close to target
- $> \mathbf{1}$: more large fish than expected


Observed vs expected length composition at Fspr40
Expected curve from nf_flicc() at SPR40; dashed line shows Lspr threshold (thresh = 0.75)
Trawl
Gillnet
2022
2023
2024
Lspr = 22.0
Obs = 0.18
Exp = 0.30
Lspr = 37.0
Obs = 0.20
Exp = 0.28
Lspr = 22.0
Obs = 0.22
Exp = 0.30
Lspr = 37.0
Obs = 0.30
Exp = 0.28
Lspr = 22.0
Obs = 0.14
Exp = 0.30
Lspr = 37.0
Obs = 0.26
Exp = 0.28
Proportion
0.00
0.05
0.10
20
40
60
Length

LBIspr in practice

Example from FLicc



- `LBIspr` tracks the upper tail of the length composition in a

LBIspr and CPUE

Complement or substitute

Potential roles of LBIspr:

1. **Corroborate CPUE**
   - if both indicators move in the same direction, confidence improves
2. **Proxy for relative abundance / exploitation signal**
   - especially in long-lived species where large fish are informative
3. **Support interpretation of selectivity and structure changes**

- `LBIspr` should not be treated as identical to CPUE.
- It is better viewed as a **structure-based index with CPUE-like utility**.

MSE applications

From assessment output to management testing

Length-based indicators can be embedded directly in **Management Strategy Evaluation (MSE)**.
Examples:

- `LBSPR` / `FLicc` outputs can support **rule-based harvest control rules**
- candidate control variables may include:
  - **SPR** or relative SPR-to-target
  - **LBIspr**
  - **mean length** L_{mean}
  - empirical ratios such as L_{mean}/L_M

This is particularly attractive for **data-limited fisheries**.

ICES Category 2 and 3 context

Link to empirical advice rules

- Under **ICES Category 2 and 3** guidelines, empirical and indicator-based approaches can be used where full analytical assessments are not available.
- This creates a natural entry point for:
  - **length-based indicators**, and
  - **SPR-based control variables**

A useful applied route is the **rule-based framework discussed by Simon Fischer**, where fishing pressure or catch advice may be linked to length-based indicators.

Example MSE note

Rule-based harvest control rules

Candidate control variables in MSE could include:

- **SPR-based F proxies** from `LBSPR` / `FLicc`
- empirical indicators such as:
  - L_{mean}
  - L_{mean}/L_M
- `LBIspr` as a structure-based abundance / exploitation proxy

Conceptually, an MSE could test whether `LBIspr`:

- tracks stock status sufficiently well,
- stabilizes advice when CPUE is poor,
- and improves robustness of empirical harvest rules.

GitHub and reproducibility

Reproducible workflow

- Repository: **github.com/Henning-Winker/FLicc**
- Includes:
  - worked examples
  - test scripts
  - equilibrium and plotting functions
  - FLR integration

A central aim is to provide a **reproducible extension of LBSPR and fishblicc concepts** within an FLR/TMB workflow.

Take-home messages

Summary

- `FLicc` extends **fishblicc** into a fast **FLR/TMB** workflow.
- It provides a reproducible bridge to **LBSPR** under the single-gear logistic special case.
- Its main strength is realistic inference for **multi-gear fisheries**.
- `LBIspr` provides a promising **structure-based indicator** that can:
  - complement CPUE,
  - substitute for CPUE when needed,
  - and be tested in **MSE / empirical HCR frameworks**.

References

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